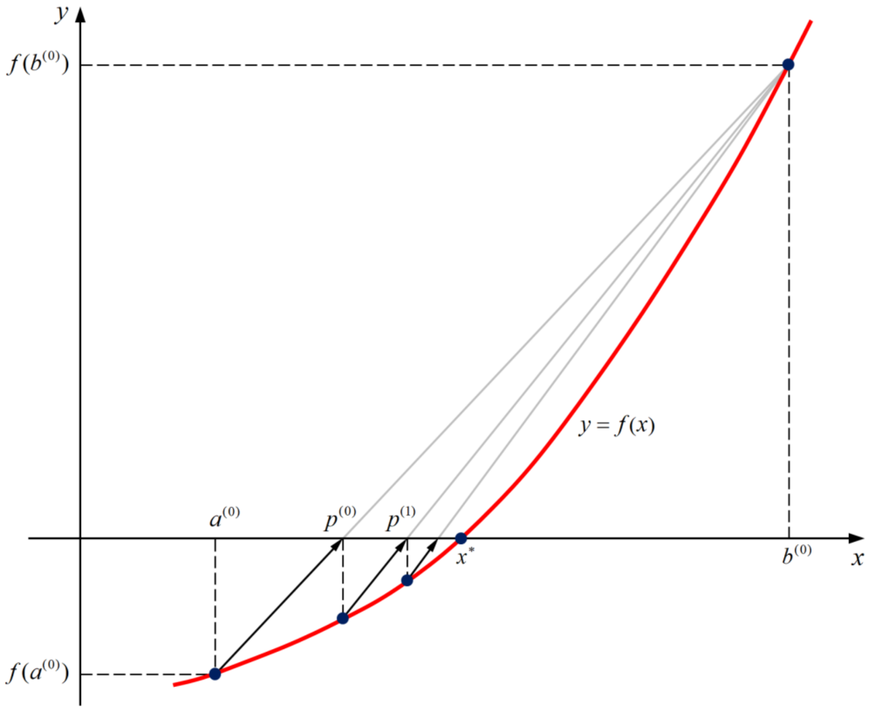
False Position Method



Bracketing Methods: There is a root, x^*, between a and b.

$$f(x^*) = 0, a < x^* < b \quad \rightarrow \quad f(a) \cdot f(b) < 0$$

False Position Method: $$p = \frac{a \cdot f(b) - b \cdot f(a)}{f(b) - f(a)}$$

$$f(a) \cdot f(p) \leq 0 \quad \rightarrow \quad a < x^* \leq p \quad \rightarrow \quad \begin{cases} a^{(k)} = a^{(k-1)} \\ b^{(k)} = p^{(k-1)} \end{cases}$$

$$f(a) \cdot f(p) > 0 \quad \rightarrow \quad p < x^* \leq b \quad \rightarrow \quad \begin{cases} a^{(k)} = p^{(k-1)} \\ b^{(k)} = b^{(k-1)} \end{cases}$$

Relative Error: $$\varepsilon^{(k)} = \frac{\left|p^{(k)} - p^{(k-1)}\right|}{\left|p^{(k)}\right|}$$

Relative Residual: $$\varepsilon^{(k)} = \frac{\left|r^{(k)}\right|}{\left|r^{(0)}\right|} = \frac{\left|f(p^{(k)})\right|}{\left|f(p^{(0)})\right|}$$

False Position Method

<u>Example 02-02</u>: Find a root of equation $x^3 + 4x^2 - 10 = 0$ in the interval $[1,2]$ using false position method with $\varepsilon_t = 10^{-6}$ for relative error.

$$a = 1, f(a) = -5$$
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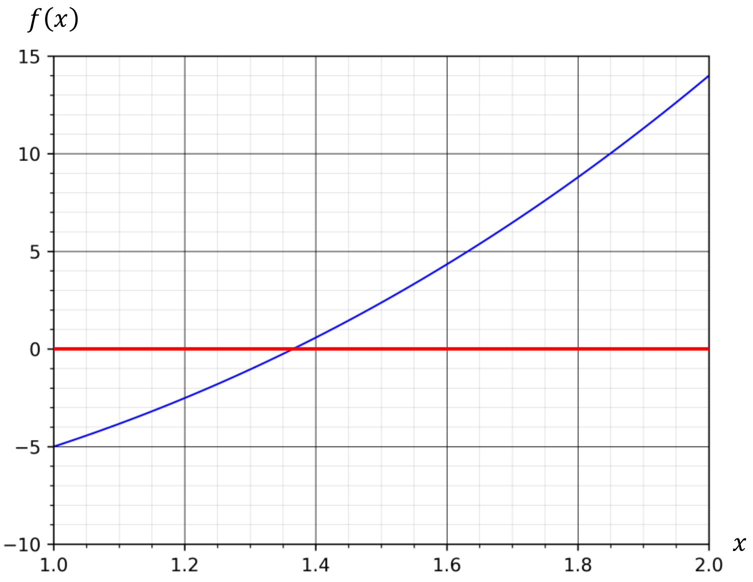

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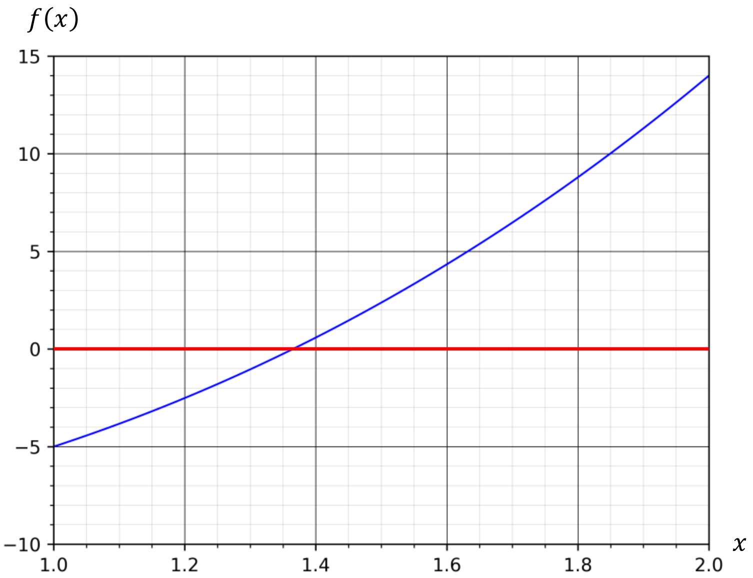

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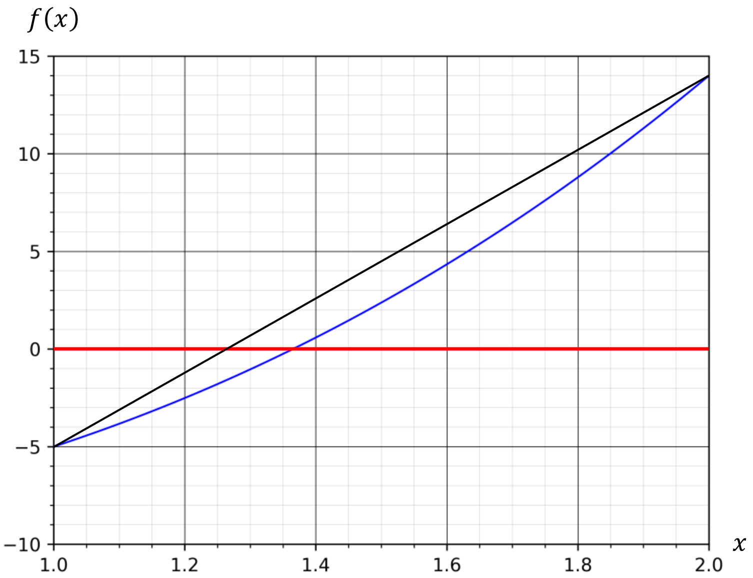

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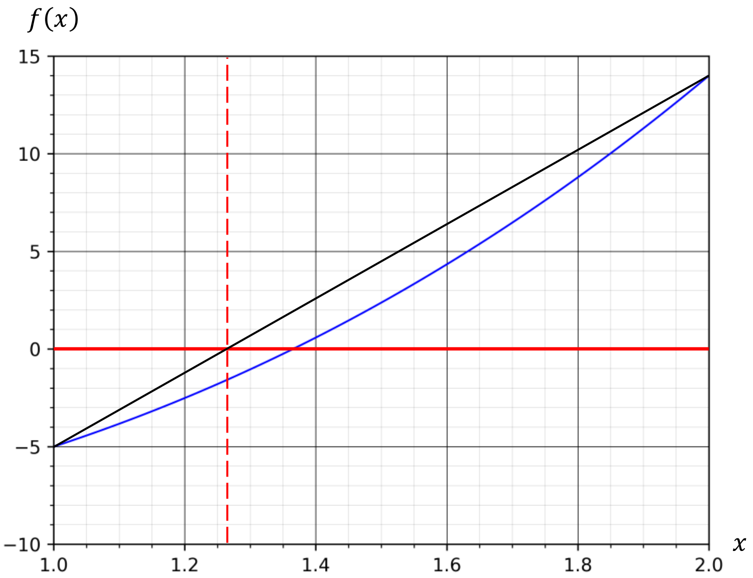

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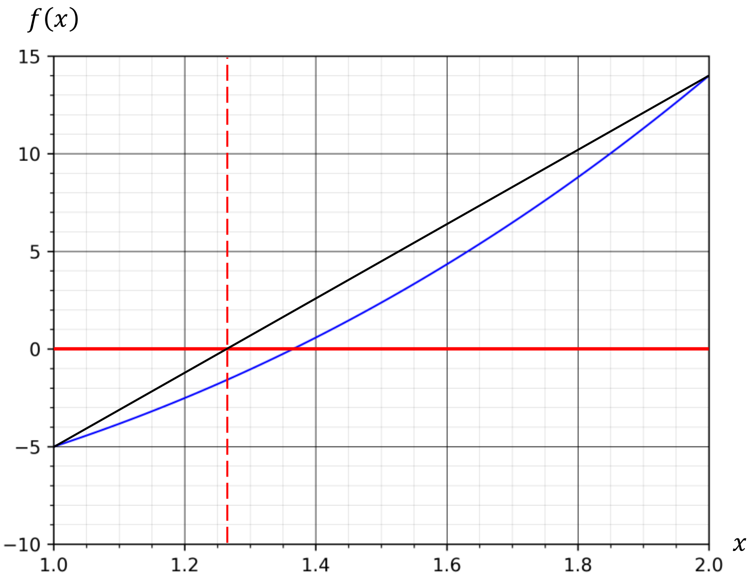

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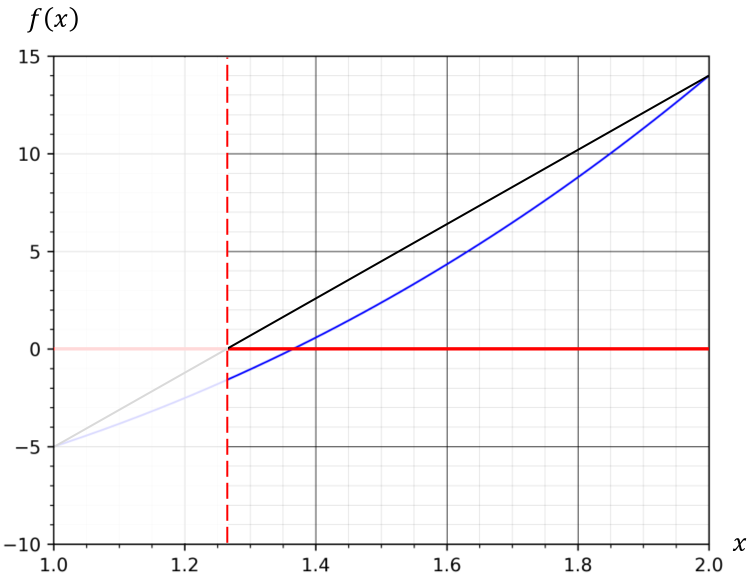

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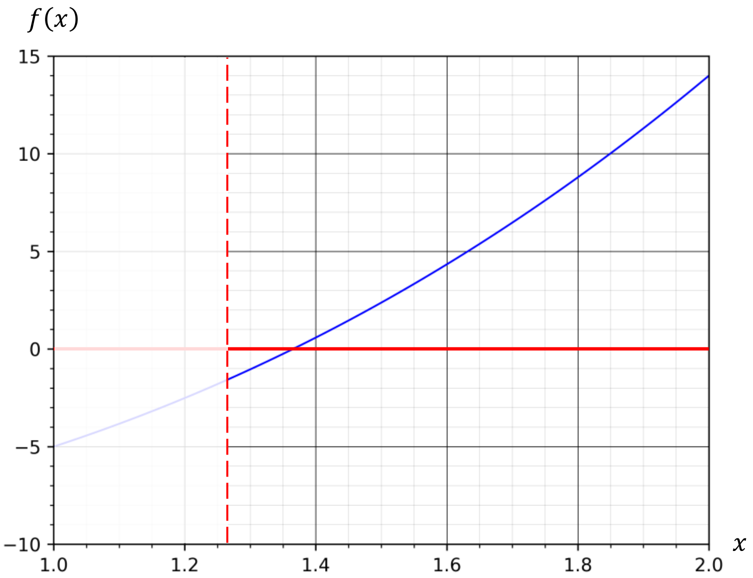

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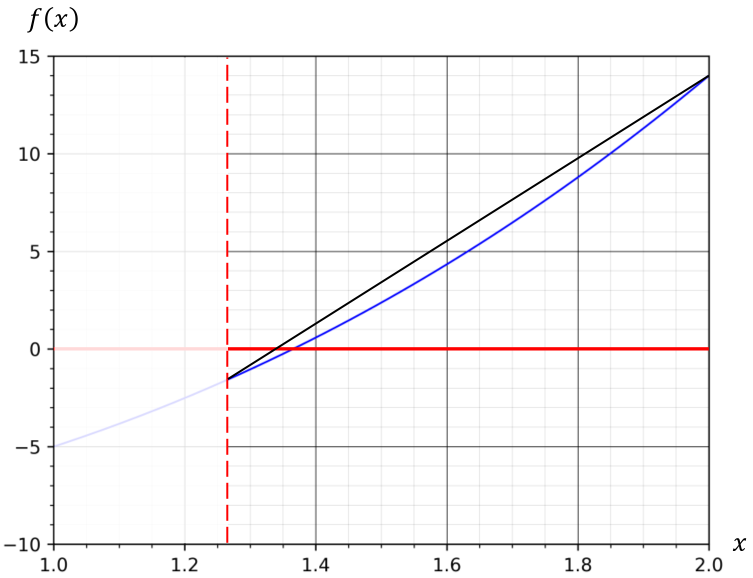

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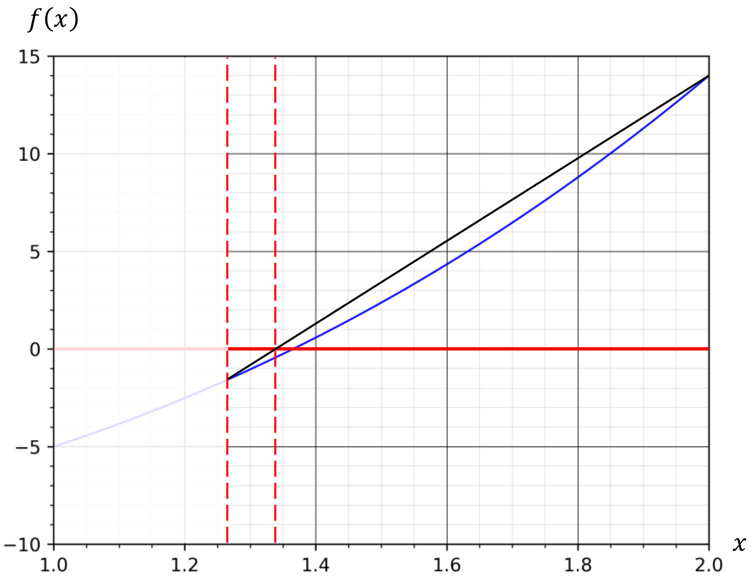

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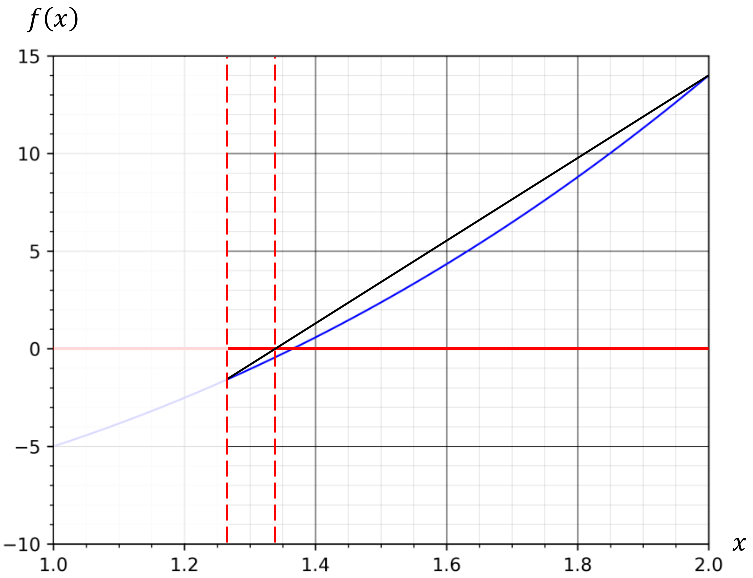

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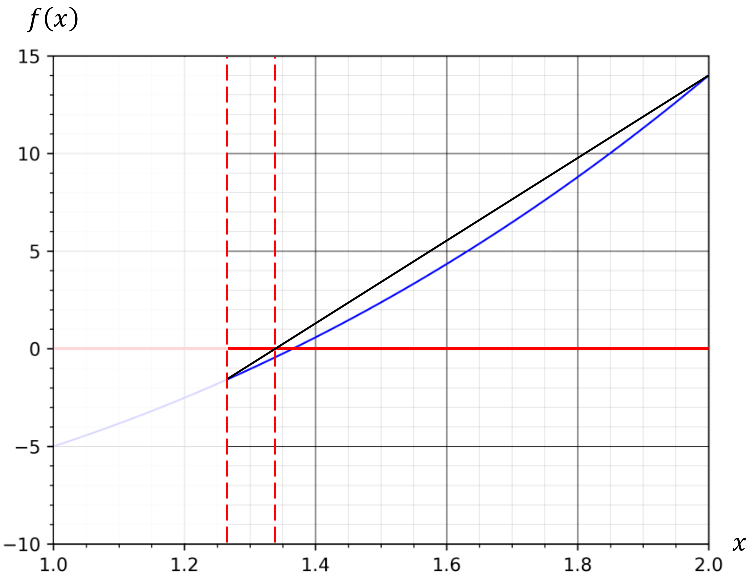

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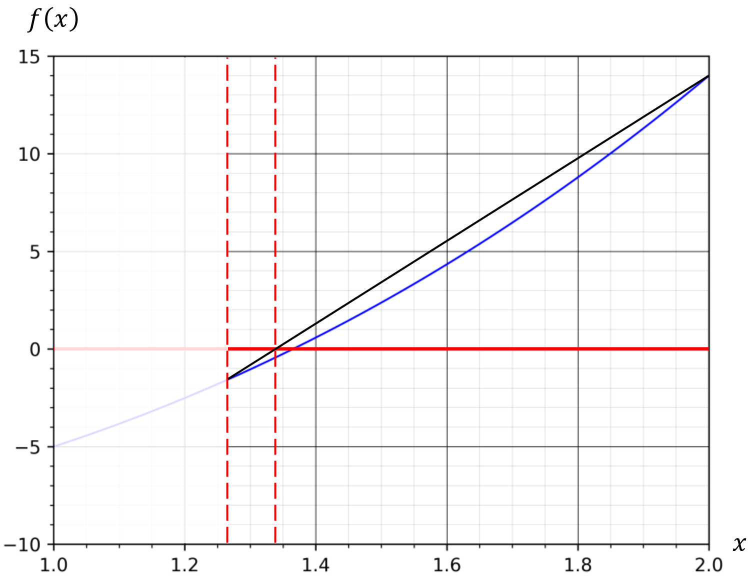

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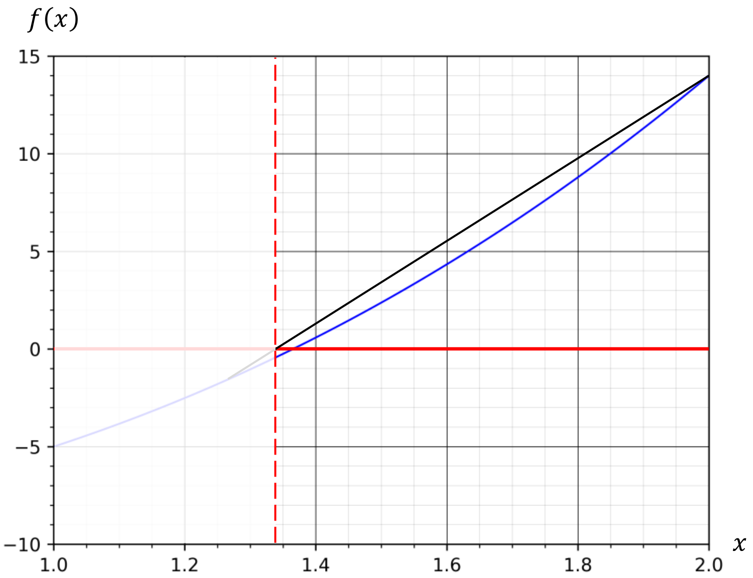

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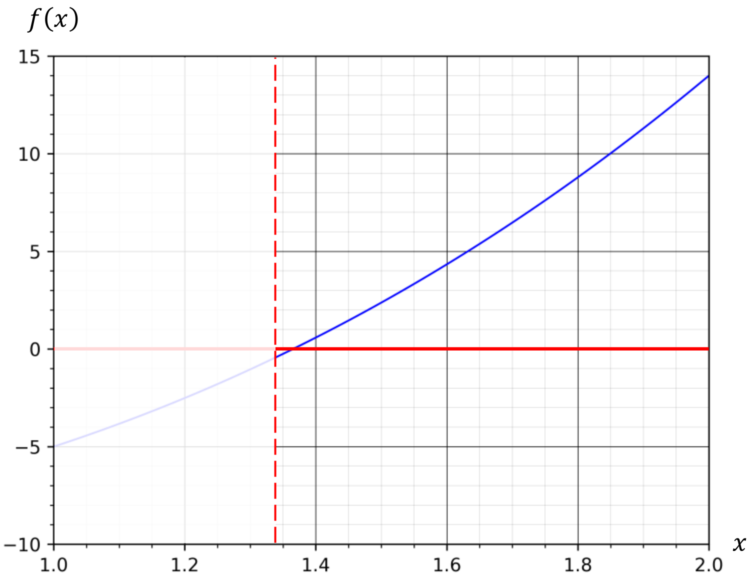

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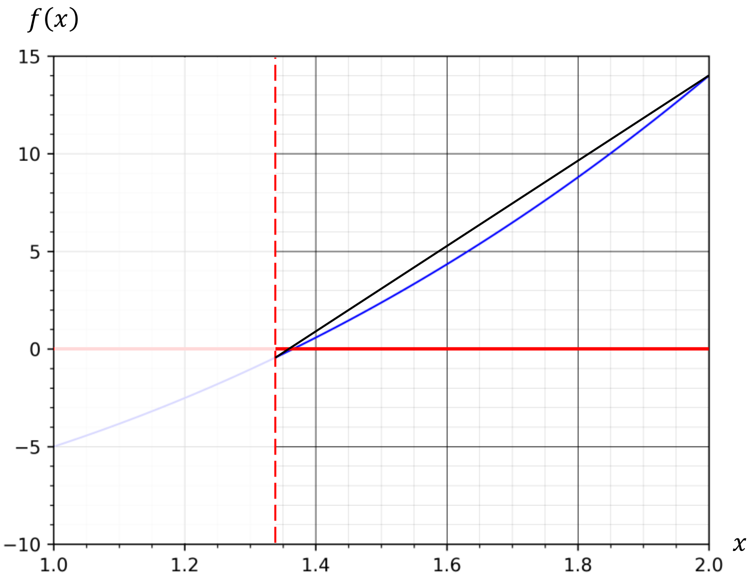

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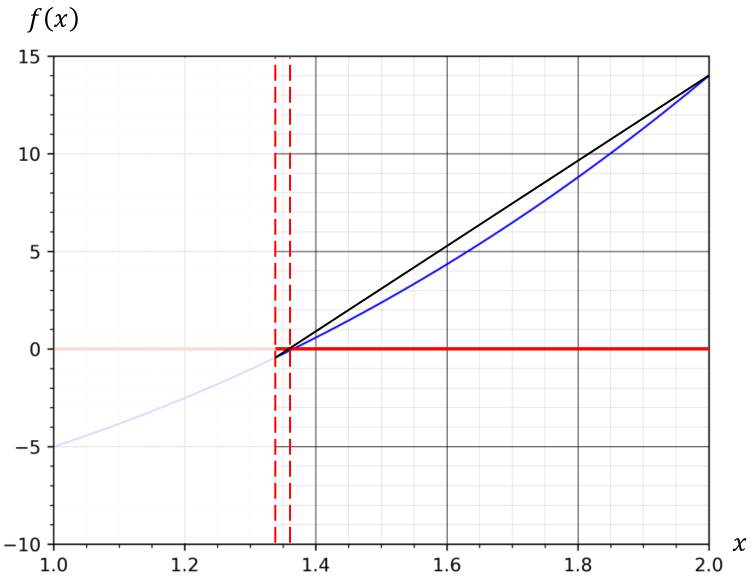

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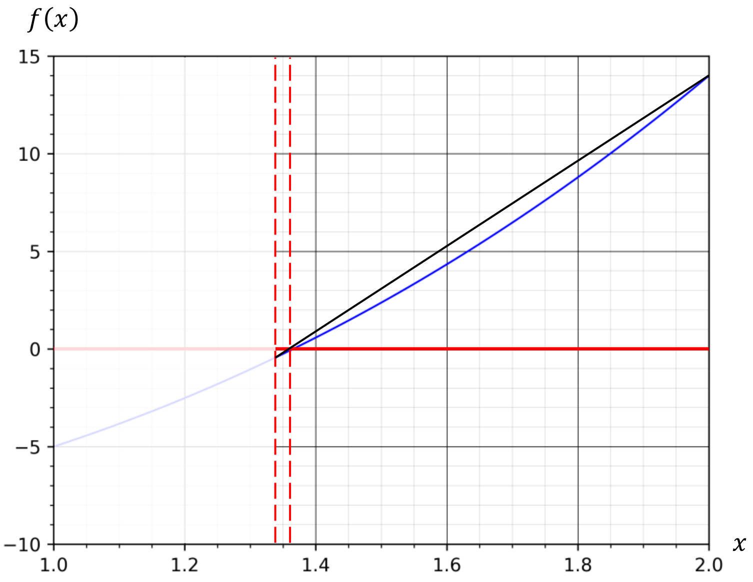

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